Optional Done Right

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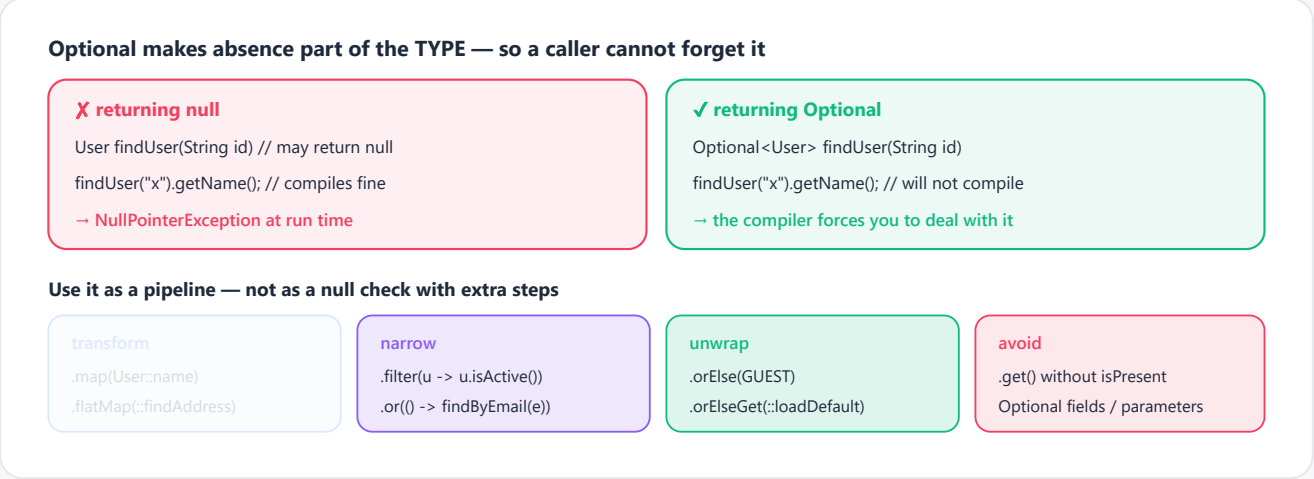
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Why it matters

`Optional` is the most misused class Java 8 added. Most code that uses it gets no benefit — `if (opt.isPresent()) opt.get()` is a null check with more syntax — and a lot of code puts it where it actively causes harm, in fields and parameters. Interviewers ask because the correct answer requires knowing what it was designed for, which is narrower than most people assume.

The concept, simply

`Optional<T>` is a container that holds either a value or nothing. Its purpose is to make absence **part of the type**, so a caller cannot silently forget it.



The value is entirely at the call site: with null, forgetting to check compiles fine. With Optional, it does not.

Brian Goetz, who led the Java 8 language work, has stated the intent plainly: `Optional` was designed as a **return type** for methods that might not return a result. It was not designed to be a general-purpose Maybe type, and specifically not for fields or parameters.

Use it as a pipeline, not as a wrapper you immediately unwrap.

Method	Purpose
`map(fn)`	Transform the value if present
`flatMap(fn)`	Transform when the function itself returns an `Optional` — avoids
`filter(pred)`	Become empty if the predicate fails
`orElse(v)`	The value, or a default — **always evaluated**
`orElseGet(supplier)`	The value, or a lazily-computed default
`orElseThrow(supplier)`	The value, or throw
`ifPresentOrElse(a, b)`	Two-branch action (Java 9+)
`or(supplier)`	Fall back to another `Optional` (Java 9+)
`stream()`	Zero-or-one stream, for flat-mapping (Java 9+)

The distinction that catches people: `orElse` **always evaluates its argument**, even when the value is present. If the default is expensive or has side effects, use `orElseGet`.

Code examples

The anti-pattern and its replacement:

JAVA

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```java
// ✗ A null check wearing a costume. No benefit whatsoever.
Optional<User> opt = findUser(id);
if (opt.isPresent()) {
    return opt.get().name();
```

```
} else {
}

// ✓ One expression, and .get() never appears.
return findUser(id).map(User::name).orElse("unknown");

// Chaining across several optional steps — flatMap avoids Optional<Option
return findUser(id)
        .flatMap(User::address)          // address() returns Optional<Add
        .map(Address::city)
        .filter(c -> !c.isBlank())
        .orElse("unknown");
```

The eager-default trap:

JAVA Copy

```
// ✗ createExpensiveDefault() runs EVERY time, even when a value is preser
User u = findUser(id).orElse(createExpensiveDefault());

// ✓ Only runs when actually needed.
User u = findUser(id).orElseGet(this::createExpensiveDefault);

// Same reasoning for exceptions — orElseThrow takes a supplier so the
// exception (and its stack trace) is only constructed on the failure path
User u = findUser(id).orElseThrow(() -> new UserNotFoundException(id));
User u = findUser(id).orElseThrow();     // NoSuchElementException (Java 1
```

Where `Optional` does not belong:

JAVA Copy

```
// ✗ As a FIELD. It is not Serializable, it adds an allocation per instanc
//   and it does not stop the field itself being null.
public class User {
    private Optional<String> middleName;   // don't
}

// ✓ Keep the field nullable and expose an Optional accessor.
```

```
    public Optional<String> middleName() {
        return Optional.ofNullable(middleName);
    }
}

// ✗ As a PARAMETER. Callers must now wrap every argument, and you still
//   have to null-check the Optional itself.
void process(Optional<String> name) { }        // don't
process(Optional.of("x"));                     // noisy at every call site

// ✓ Overload, or accept nullable and document it.
void process(String name) { }
void process() { }

// ✗ In a collection. An empty list already means "nothing".
List<Optional<String>> names;                  // don't
```

Genuinely good uses beyond the basic return:

JAVA Copy

```
// Two-branch handling without an if (Java 9+)
findUser(id).ifPresentOrElse(
    user -> log.info("found {}", user.name()),
    ()   -> log.warn("no user {}", id));

// Fallback chain (Java 9+)
Optional<User> user = findByEmail(e).or(() -> findByPhone(p)).or(() -> fin

// Flattening optionals inside a stream - drops the empties in one step (J
List<User> found = ids.stream()
                      .map(this::findUser)      // Stream<Optional<User>>
                      .flatMap(Optional::stream) // Stream<User>
                      .toList();

// Primitive versions avoid boxing where it matters:
OptionalInt max = IntStream.of(1, 2, 3).max();
```

`Optional` is not `Serializable`, which is the concrete reason it must not be a field on anything that gets serialised — a JPA entity, a session object, a DTO going over RMI. It will simply fail. It also costs an allocation per instance, which is why it does not appear in performance-critical JDK APIs.

`Optional.of` versus `ofNullable` is a small but real distinction: `of` throws `NullPointerException` if handed null, which is correct when null would indicate a bug; `ofNullable` converts null into an empty `Optional`, which is what you want at a boundary with legacy code. Using `of` where the value may legitimately be null is a common mistake that turns a graceful absence into a crash.

Never call `get()` without a check. It throws `NoSuchElementException`, and it defeats the point of the type. Java 10 added `orElseThrow()` with no arguments as a better-named equivalent, and modern style is to treat plain `get()` as deprecated in spirit even though it is not formally.

Where the JDK itself uses it is instructive: `Stream.findFirst`, `Stream.max`, `Optional`-returning lookups. All of them are return values from operations that may legitimately produce nothing. That is the whole design space.

Common mistakes

- **`isPresent()` then `get()`.** A null check with extra ceremony. Use `map` / `orElse`.
- **Calling `get()` unguarded.** Throws, and defeats the type.
- **`Optional` as a field.** Not serialisable, an extra allocation, and the field can still be null.
- **`Optional` as a parameter.** Forces every caller to wrap. Overload instead.
- **`orElse` with an expensive default.** It is evaluated even when unnecessary — use `orElseGet`.
- **Returning `null` from a method declared to return `Optional`.** The worst of both worlds, and it happens.
- **`Optional` inside collections.** An empty collection already expresses absence.
- **`Optional.of` on a possibly-null value.** Throws. Use `ofNullable`.

Q: "What is `Optional` for?"

A: Making absence explicit in a method's return type, so callers cannot silently forget the empty case — the compiler makes them deal with it. Brian Goetz has said directly that it was designed as a return type for methods that might not have a result, not as a general Maybe type, which is why using it for fields and parameters goes against the design.

Q: "What is wrong with `isPresent()` and `get()`?"

A: It is a null check with more syntax and no added safety — you have wrapped the value only to immediately unwrap it. The point is to compose: `map`, `flatMap`, `filter`, then `orElse` or `orElseThrow` at the end. If `get()` appears in my code I treat it as a sign the pipeline was not finished.

Q: "`orElse` or `orElseGet`?"

A: `orElse` takes a value, so the argument is evaluated whether or not it is needed. `orElseGet` takes a supplier and only runs on the empty path. For a constant either is fine; for anything expensive or with side effects — a database lookup, object construction — `orElseGet` is required, and the same reasoning is why `orElseThrow` takes a supplier.

Q: "Should `Optional` be used for fields or parameters?"

A: No to both. As a field it is not serialisable, costs an allocation per instance, and does not stop the field itself being null — better to keep the field nullable and return `Optional.ofNullable( ... )` from the getter. As a parameter it forces every caller to wrap their argument and you still have to null-check the `Optional`; overloaded methods express optional arguments better.

A: `map` when the function returns a plain value; `flatMap` when it already returns an `Optional`, so you get `Optional<T>` rather than `Optional<Optional<T>>`. It is the same relationship as in streams — chaining lookups that each may fail is exactly what `flatMap` is for.

Q: "Does `Optional` replace null?"

A: No, and treating it that way is where the misuse comes from. It is a return-type idiom for "this might not produce a result". Nulls still exist in fields, in collections and across library boundaries, and wrapping everything in `Optional` adds allocation and noise without making those safer. For the rest, `Objects.requireNonNull` at boundaries and helpful NullPointerExceptions from Java 14 do more good.

Cheat-card

- Purpose: **a return type** for "might not produce a result". Not a general Maybe.
- **Never** a field (not `Serializable`), a parameter, or a collection element.
- Compose: `map` → `flatMap` → `filter` → `orElse` / `orElseThrow`. **`get()` should never appear.**
- `orElse` is **eager**; `orElseGet` / `orElseThrow` are lazy.
- `of` throws on null; `ofNullable` converts it to empty.
- Java 9+: `ifPresentOrElse` · `or` · `Optional::stream` for flat-mapping.
- Primitives: `OptionalInt` / `OptionalLong` / `OptionalDouble`.
- Never return `null` from a method declared `Optional`.

QUICK CHECK — no marks, just for you.

1. What was Optional designed for?

- ◯ Wrapping fields so they can be nullable

- Replacing null everywhere in the codebase
- A return type for methods that might not produce a result, so callers cannot silently ignore the empty case

Check answer

2. What is the difference between orElse and orElseGet?

- orElse takes a VALUE and always evaluates it, even when a value is present; orElseGet takes a supplier and only runs on the empty path
- They are identical
- orElseGet throws if the Optional is empty
- orElse is for objects and orElseGet is for primitives

Check answer

3. Why should Optional not be used as a field type?

- It always throws when serialised to JSON
- It is not Serializable, it adds an allocation per instance, and it does not even stop the field itself being null
- Fields cannot hold interfaces
- It cannot be used with generics

Check answer

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Hand-picked references for this lesson.